

Real-World Effectiveness of Dapagliflozin and Sitagliptin Fixed Dose Combination in Indian Patients with Type 2 Diabetes: A Retrospective Analysis of Electronic Medical Records Stratified by BMI

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INTRODUCTION & OBJECTIVE

Burden

The incidence of T2DM is rapidly increasing resulting in significant diabetes-related complications ¹

Global prevalence ^{1,2}

537 million

Increased by 90.5% from 1990 and 2021

Indian prevalence ^{3,4}

101 million

Increased from 5.5% in 1990 to 9.6% in 2021

- More than 75% of Indian patients have uncontrolled diabetes (HbA1c $\geq 7\%$), requiring sequential addition of medications with multiple tablets ^{5,6}
- Approximately 80 to 90% of patients with T2DM are either overweight or obese which can further cause complications, such as cardiovascular disease ^{7,8}
- The prevalence of diabetes increases with increasing BMI levels. For instance, the prevalence of diabetes ranged from 2.4% in underweight individuals to 30.4% among obese individuals ⁹

Importance of FDC

Combination therapy provides effective control of blood glucose and do not increase the risk of hypoglycemia or other AEs ⁹

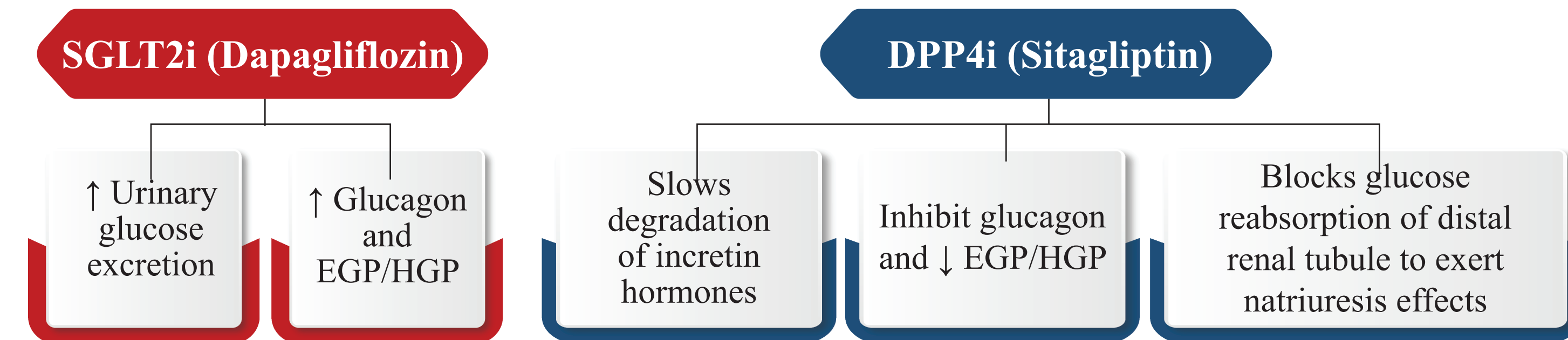
Benefits of FDC over multiple OADs as combination therapy

FDC ^{6,10}	Multiple OADs ⁶
- Reduces pill burden, dosing frequency, and cost - Improved adherence - Complementary mechanism of action - Sustained glycemic control	- Increases pill burden and dosing frequency - Poor adherence leading to inadequate glycemic control

Synergistic effect of Dapagliflozin and Sitagliptin FDC

- Dapagliflozin and Sitagliptin FDC can benefit patients with T2DM due to its ability to address multiple pathophysiologies of T2DM, potentially simplifying therapy.^{12,13,14} Additionally in Indian patients,
 - DPP4i is widely used, well-tolerated, and has increased efficacy due to an increase in DPP4 enzyme activity ^{13,15,16}
 - SGLT2i, besides glycemic control, offers unique benefits such as weight reduction, improvement in β -cell function, beneficial effect on cardiovascular and renal system¹⁴

Synergistic effect of Dapagliflozin and Sitagliptin FDC ¹¹

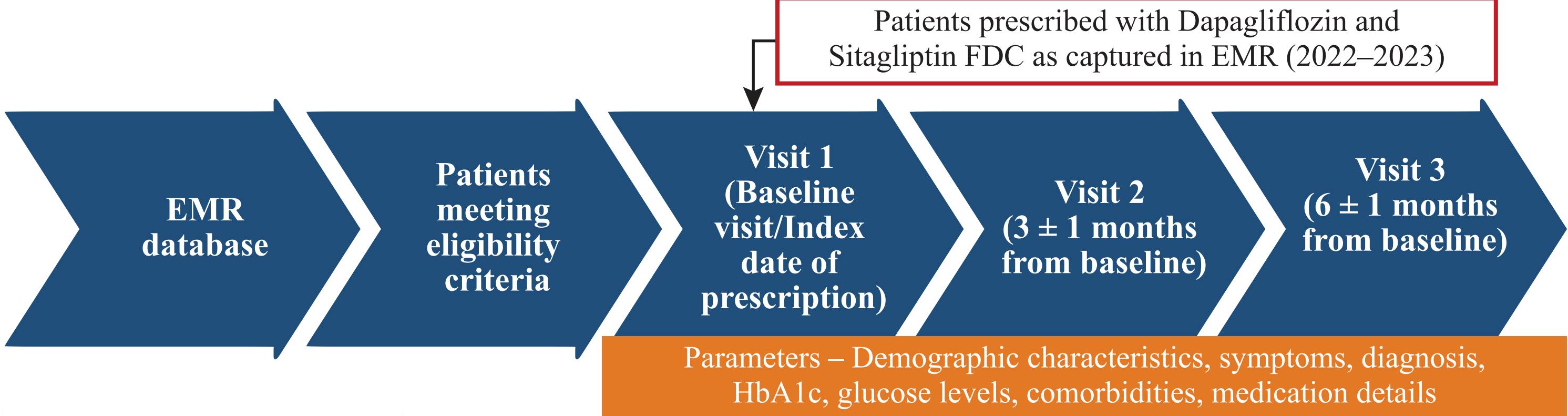


Need and objective of real-world evidence

- RWE is essential to understand the generalizability and applicability of Dapagliflozin and Sitagliptin FDC, in clinical practice and to help clinicians in making informed decisions about patient treatment ¹⁷
- Limited real-world data exist on the effectiveness of Dapagliflozin and Sitagliptin FDC in India ¹²
- This retrospective, observational, EMR-based study was conducted to assess the real-world effectiveness and usage pattern of Dapagliflozin and Sitagliptin FDC in Indian patients. **The results of BMI stratified analysis is presented here.**

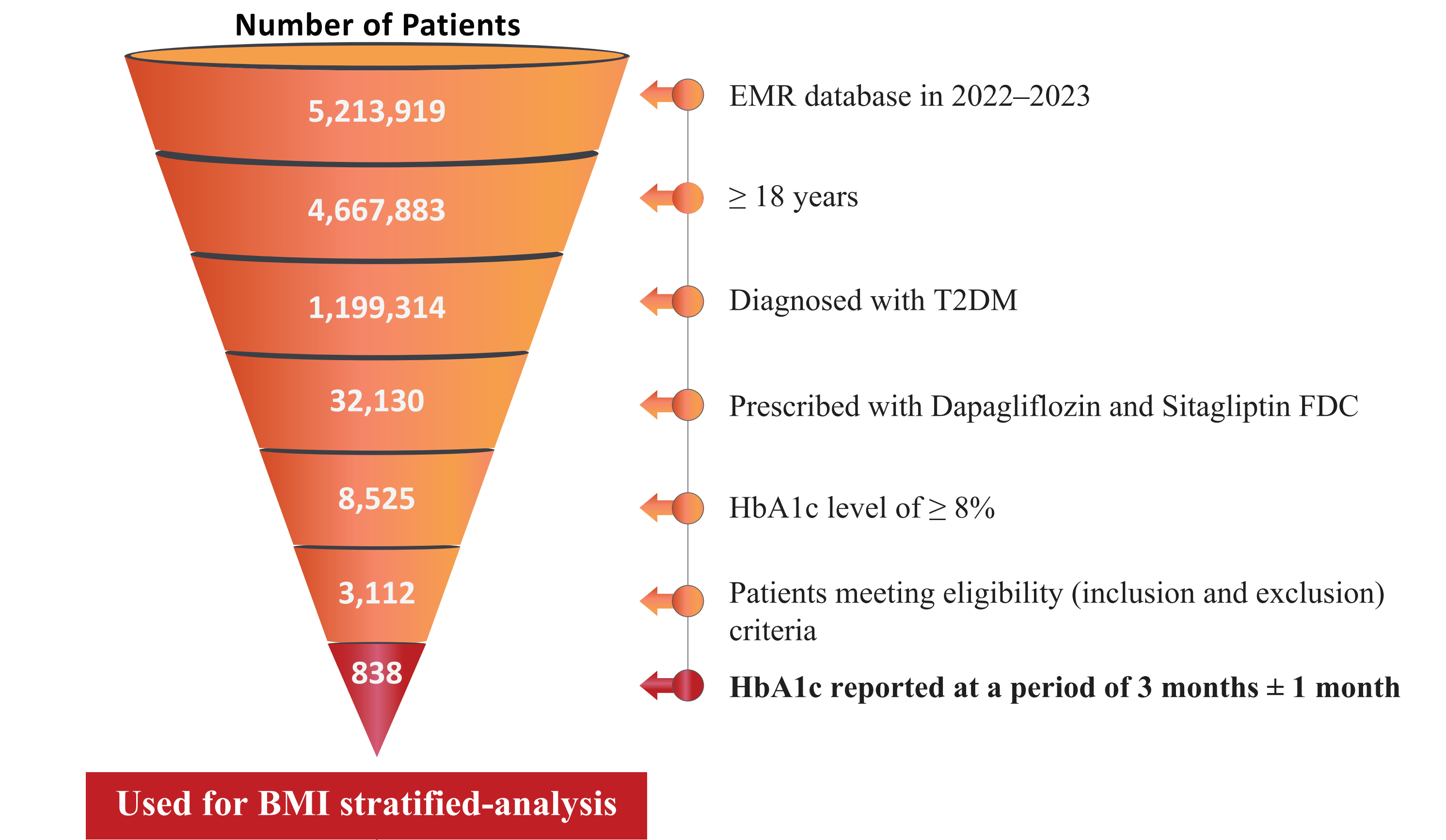
METHODS

- In this study, aggregated and anonymized EMR data from the patients meeting the eligibility criteria was retrieved and analyzed. Study was approved by Independent Ethics Committee and registered on CTRI (CTRI no. CTRI/2023/10/058365)



Inclusion criteria	Exclusion criteria	Primary endpoint	Key secondary endpoints
- Patients with - Age ≥ 18 years - Diagnosed with T2DM - HbA1c $\geq 7\%$ - Prescribed Dapagliflozin and Sitagliptin FDC in any visit other than baseline visit on the EMR platform - EMR data captured for more than one visit	- T1DM - Patients on insulin or any other injectable antidiabetic medication - Patients with incomplete medical records with respect to the required study parameters	Mean change in HbA1c from baseline to 3 months in patients with HbA1c $\geq 8\%$ at baseline	- Mean change in FBG and PPBG level from baseline to 3 months in patients with HbA1c $\geq 8\%$ at baseline - Mean change in body weight from baseline to 3 months in patients with HbA1c $> 8\%$ with BMI > 23 kg/m² and > 25 kg/m² at baseline

RESULTS



- Group 1: Normal (18.50-22.99 kg/m²; N = 41)
 - Group 2: Overweight (23-24.99 kg/m²; N = 64)
 - Group 3: Obese (≥ 25 kg/m²; N = 361)
- Mean overall reduction in HbA1c reduced by 1.31% from 9.29% at baseline to 7.98% at 3 months ($p < 0.001$; Figure 1)
 - A significant reduction in mean HbA1c levels was observed across all BMI groups from baseline to 3 months ($p < 0.001$, Figure 1)

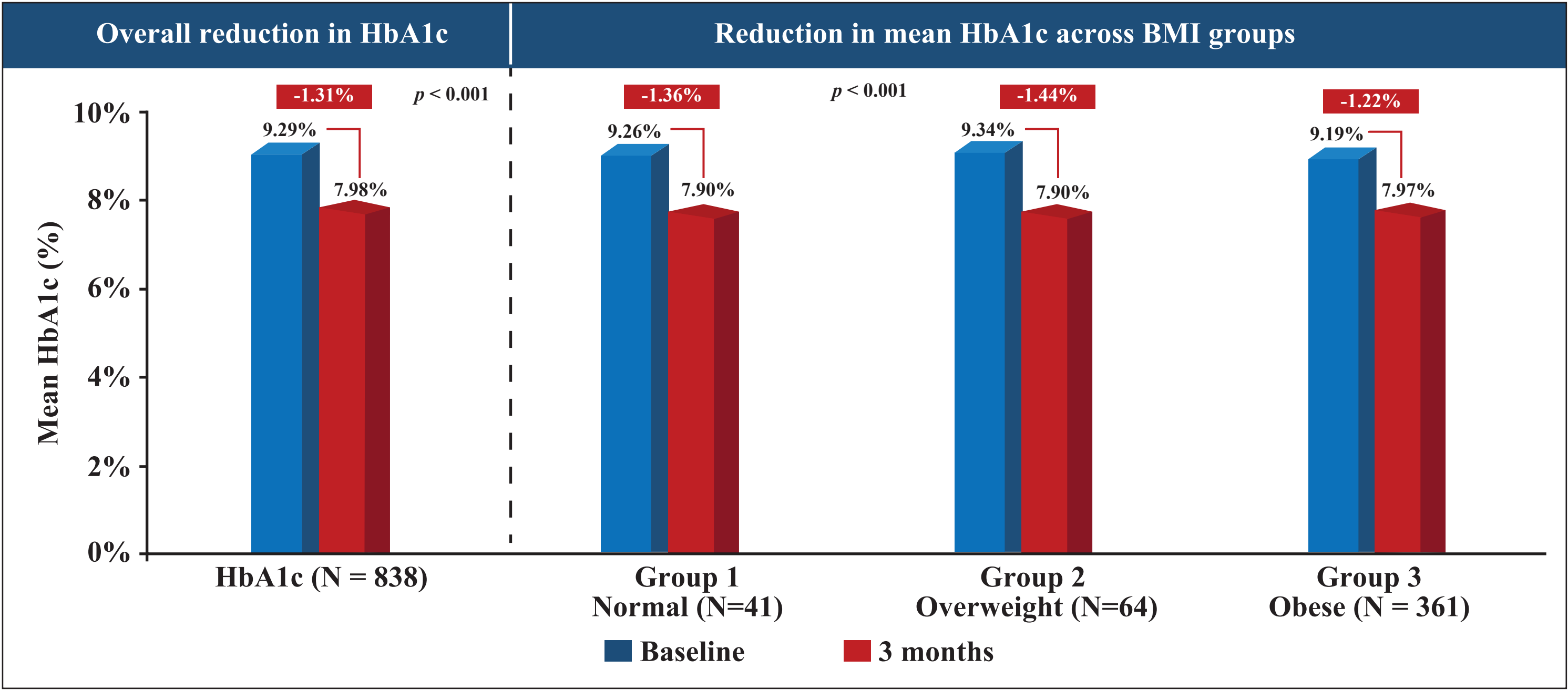


Figure 1: Mean change in HbA1c from baseline to 3 months and among different BMI groups

- Significant reduction in FBG and PPBG was observed in all the BMI groups from baseline to 3 months ($p < 0.001$, Table 1 and 2)

Table 1: Mean change in FBG (mg/dL) among different BMI groups			
	Group 1: Normal (N = 35)	Group 2: Overweight (N = 56)	Group 3: Obese (N = 303)
Baseline	165.21	162.85	163.91
3 months	139.78	124.73	134.17
Change (mg/dL)	-25.43	-38.1	-29.74

Table 2: Mean change in PPBG (mg/dL) among different BMI groups			
	Group 1: Normal (N = 34)	Group 2: Overweight (N = 51)	Group 3: Obese (N = 249)
Baseline	248.2	231.96	244.22
3 months	200.69	188.74	197.97
Change (mg/dL)	-47.55	-43.22	-46.25

- Significant reduction in mean body weight was observed in patients with BMI > 23 kg/m² and > 25 kg/m² from baseline to 3 months and 6 months ($p < 0.001$, Figure 2)

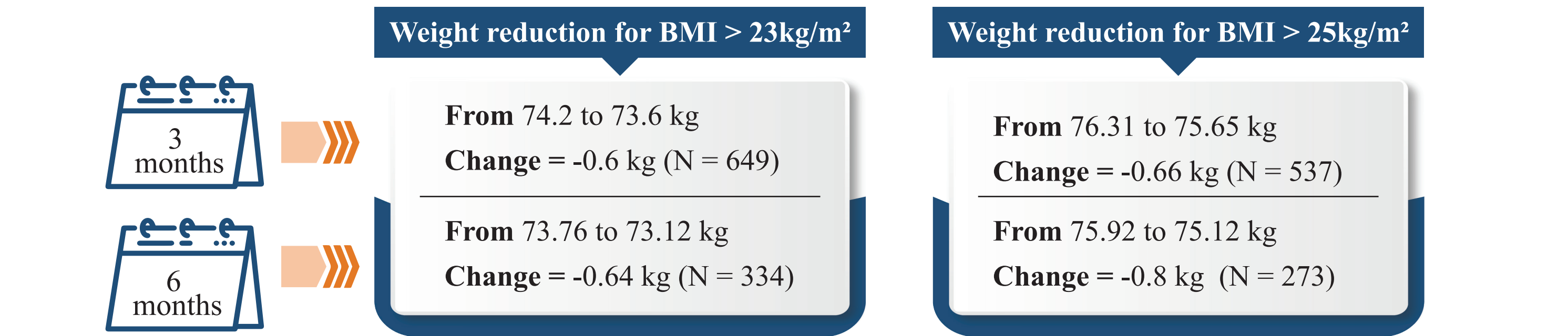


Figure 2: Mean change in body weight among different BMI groups

CONCLUSION

- Dapagliflozin and Sitagliptin FDC was effective in overall glycemic control in patients with T2DM.
- Dapagliflozin and Sitagliptin FDC effectively reduced HbA1c in patients with T2DM across all BMI groups within 3 months, indicating effective glycemic control.
- Both FBG and PPBG levels decreased significantly within each BMI stratum, emphasizing the effectiveness of the intervention in controlling blood sugar levels.
- In addition, the FDC also significantly reduced body weight highlighting its potential in managing both, glycemic control and weight in patients with T2DM.
- Results from this subgroup analysis suggest that Dapagliflozin and Sitagliptin FDC is an effective treatment option across different BMI with T2DM.

Disclosure of interest: Ritesh Agrawala: None declared, Supratik Bhattacharyya: None declared, Prakadeesh Bharathi B: Author is full time employee of Sun Pharma Laboratories Ltd., Vinay Kudrigikar: Author is full time employee of Sun Pharma Laboratories Ltd., Prachi Ahire: Author is full time employee of Sun Pharma Laboratories Ltd., Shruti Dharmadhikari: Author is full time employee of Sun Pharma Laboratories Ltd., Chintan Khandhedia: Author is full time employee of Sun Pharma Laboratories Ltd., Neeraj Markandeywar: Author is full time employee of Sun Pharma Laboratories Ltd., Amey Mane: Author is full time employee of Sun Pharma Laboratories Ltd., Suyog Mehta: Author is full time employee of Sun Pharma Laboratories Ltd

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Abbreviations: BMI, body mass index; AE, adverse event; EGP, endogenous glucose production; EMR, electronic medical record; FDC, fixed dose combination; FBG, fasting blood glucose; HbA1c, glycated hemoglobin; HGP, hepatic glucose production; OAD, oral antidiabetic drug; PPBG, postprandial blood glucose; T1DM, type 1 diabetes mellitus; T2DM, type 2 diabetes mellitus

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